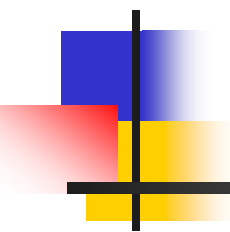


Network Model (Protocol and Layering)



Assoc. Prof. Anan Phonphoem, Ph.D. (วศ.ดร.อนันต์ ผลเพิ่ม)

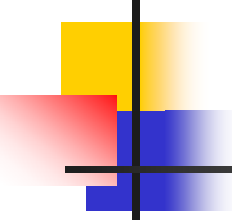
Email: anan.p@ku.ac.th <http://www.cpe.ku.ac.th/~anan>

Asst. Prof. Aphirak Jansang, D.Eng. (ผศ.ดร.อภิรักษ์ จันทร์สราง)

Email: aphirak.j@ku.ac.th <http://www.cpe.ku.ac.th/~aphirak>

Computer Engineering Department

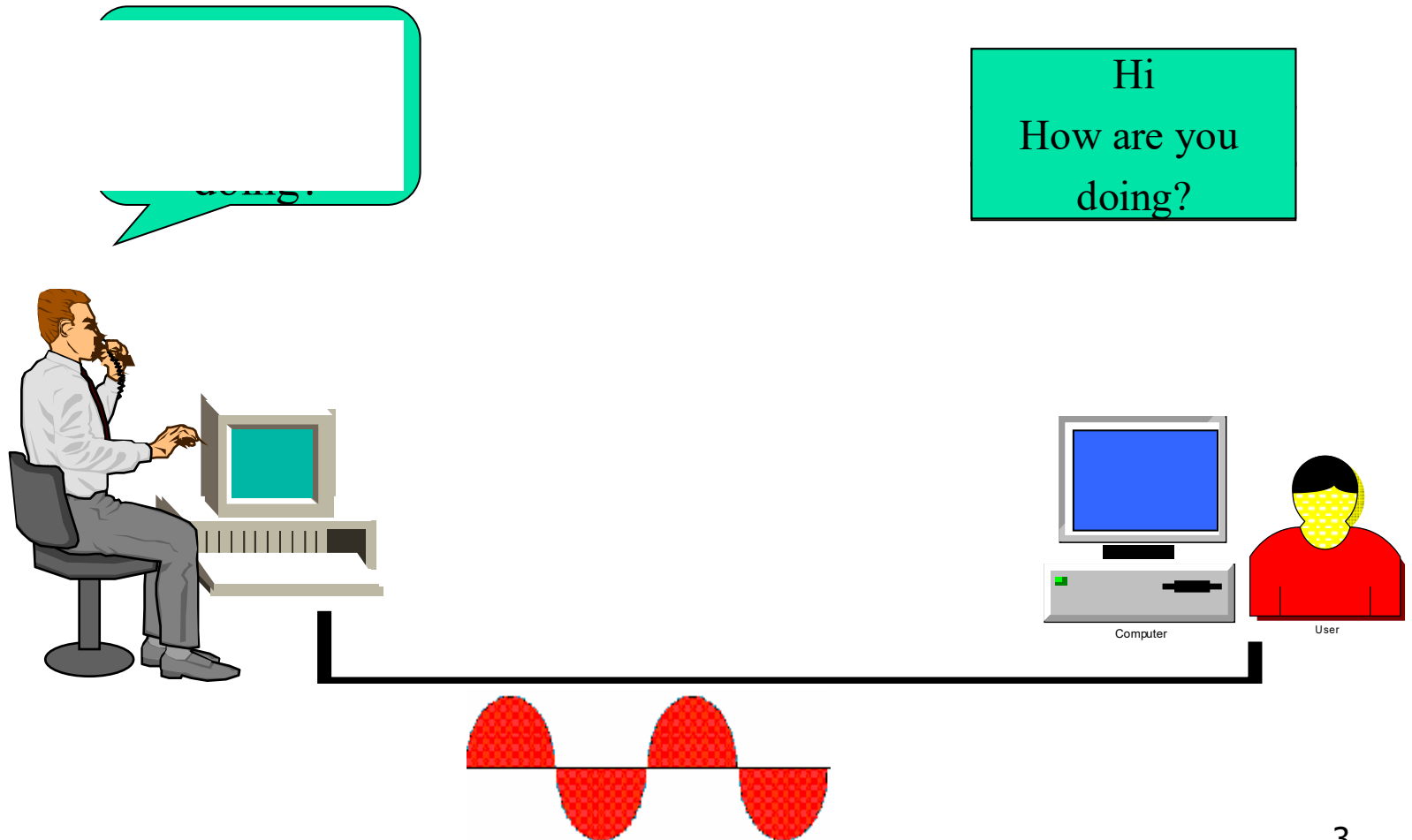
Kasetsart University, Bangkok, Thailand



Outline

- **Network Model**
- Layering Concept
- Techniques protocols use

Data Communication



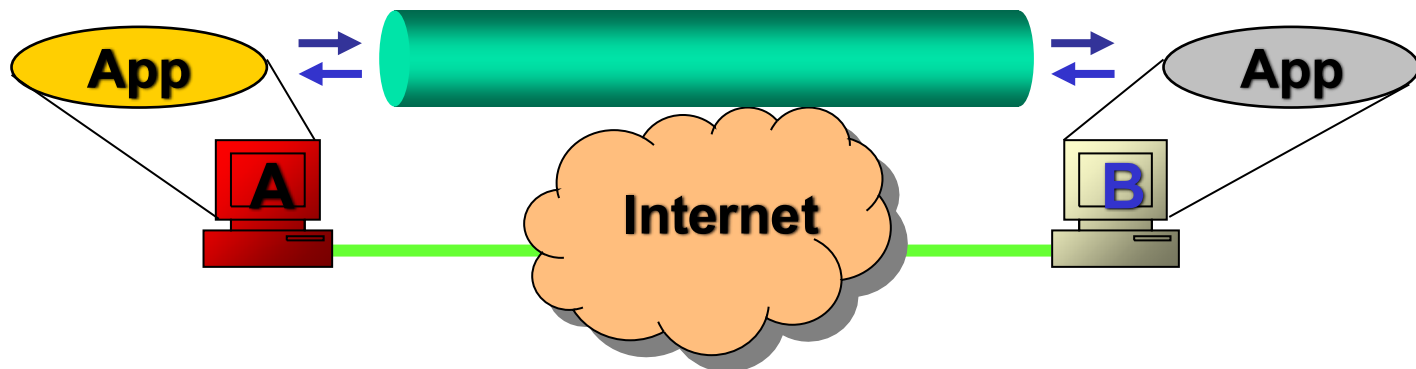


How can it happen?

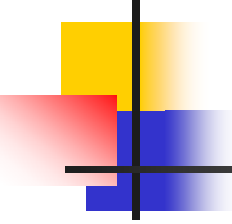
- A lot of work since user inputs data until the data is transmitted.
- Should each application program take care all tasks?
 - NO

Communication: App's Viewpoint

- Two network applications interact as if there is a dedicated pipe connecting them



- But what's going on underneath?
 - What is inside the "cloud"?



Computer Communication Model

- Modularity
- Well-defined interfaces
- Well-defined functions and protocol

Lord of the Rings



novel written by
J. R. R. Tolkien.



<http://wallpapercave.com/map-of-middle-earth-wallpaper>

Sauron attack → **Minas Tirith**

Lord of the Rings



Gandalf, The Wizard

<https://www.mirror.co.uk/>



Pippen, The Hobbit

<https://www.wattpad.com/>



Théoden, King of Rohan

wikipedia

Lord of the Rings: Lighting of the Beacons

- **Gondor** (by Gandalf) sends a **signal** to **Rohan** asking them for military aid



<https://www.youtube.com/watch?v=GDYQ3PJKrT8>

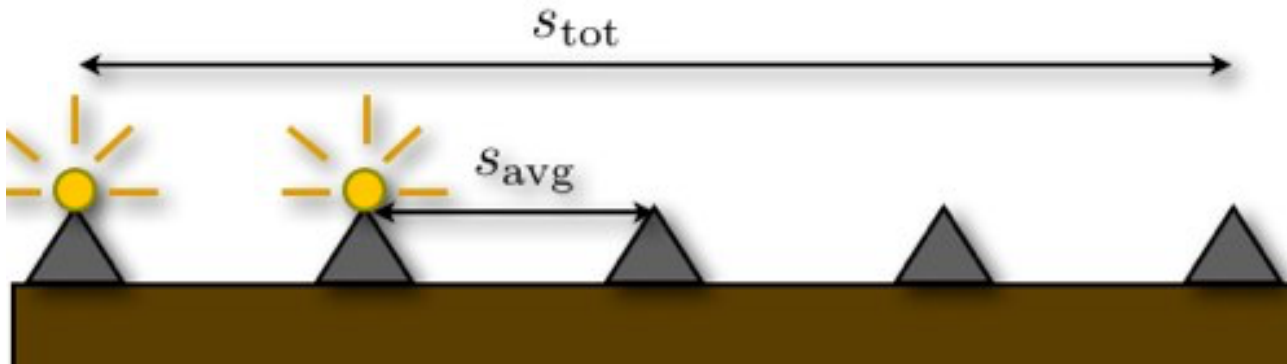
Question?



How fast is the Beacon of Gondor?

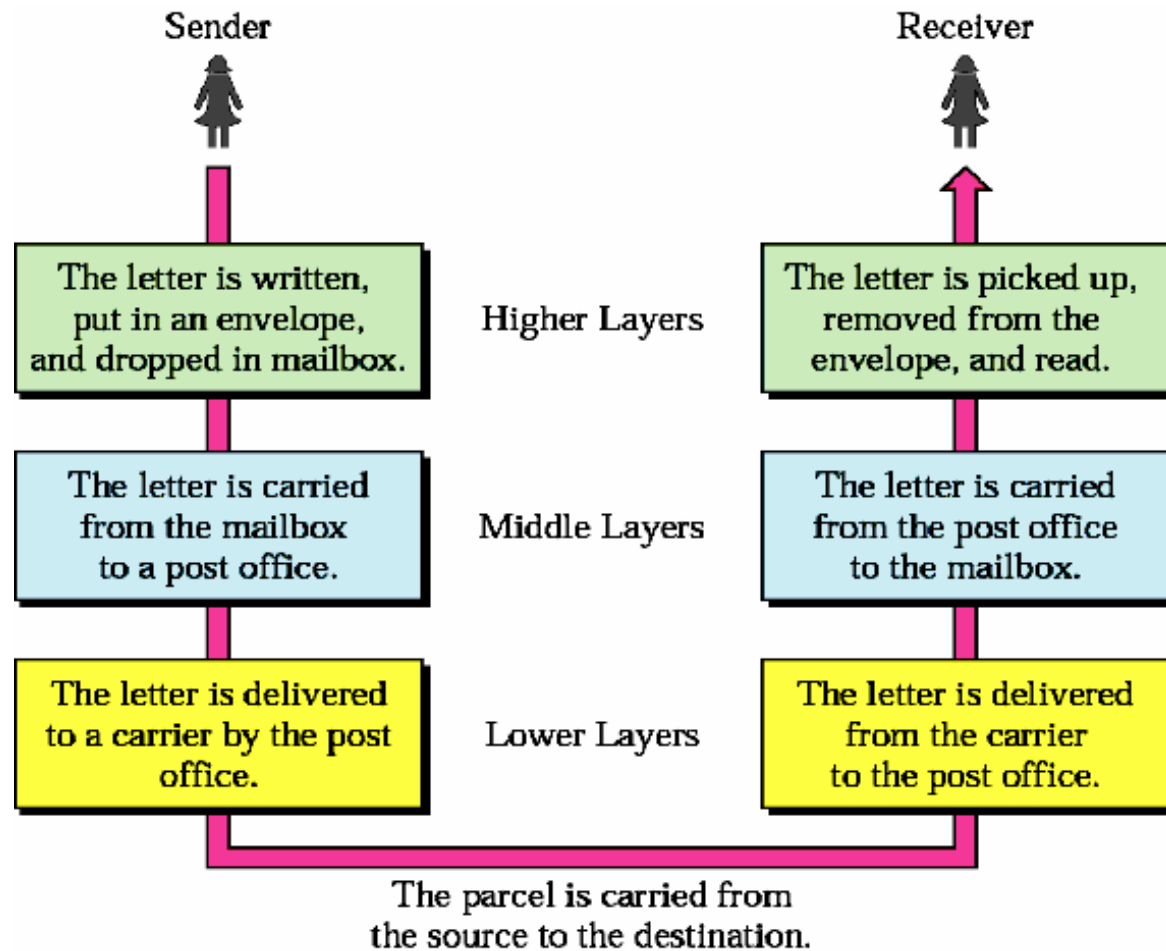
Posted by Rhett Allain on July 30, 2010

<http://scienceblogs.com/dotphysics/2010/07/30/how-fast-is-the-beacon-of-gondor/>



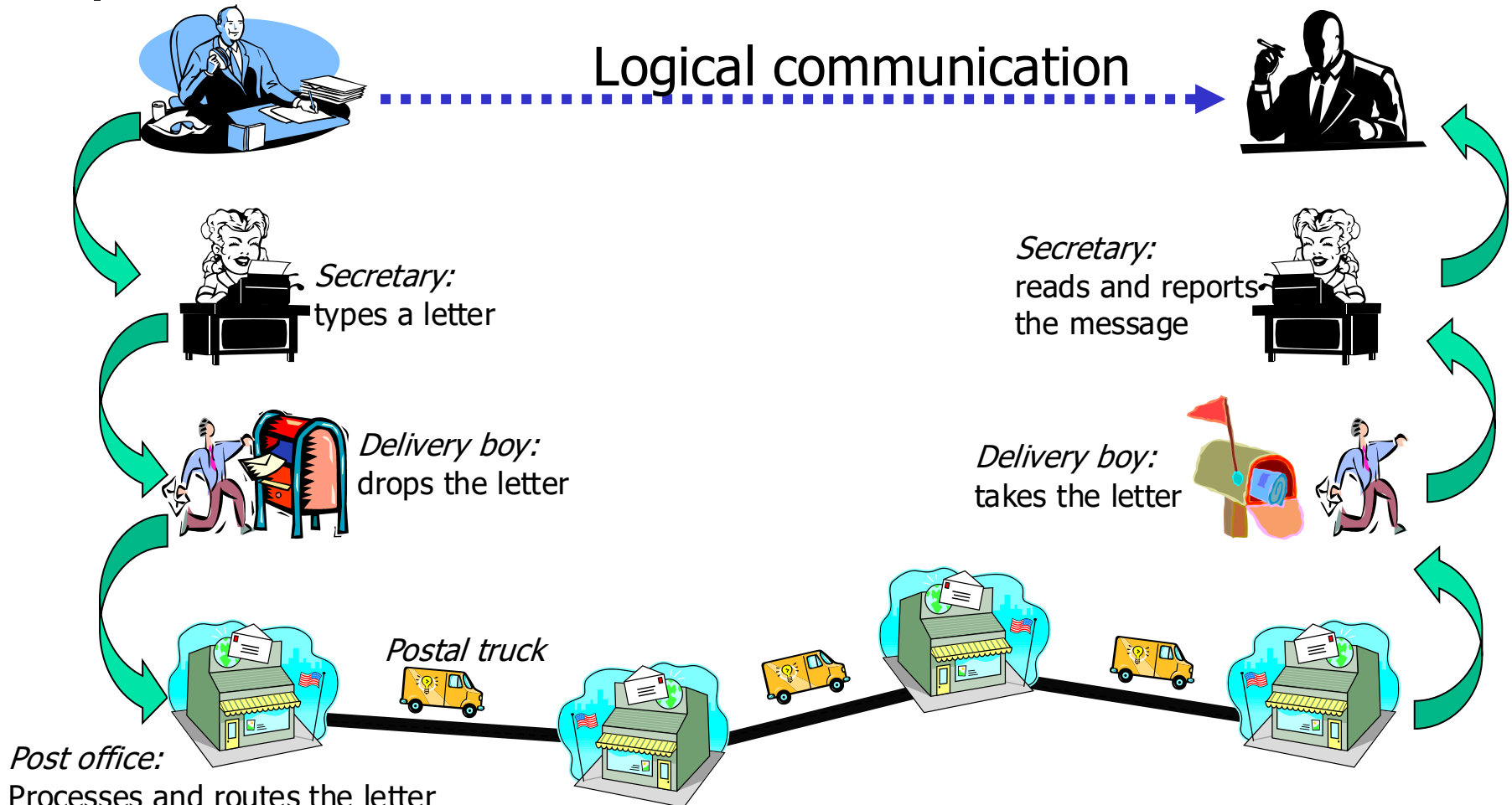
$$v_{\text{signal}} = \frac{n s_{\text{avg}}}{n(t_l + t_r)} = \frac{s_{\text{avg}}}{t_l + t_r}$$

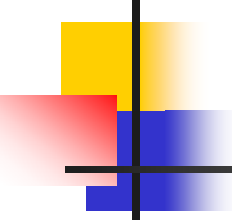
Transmit a letter



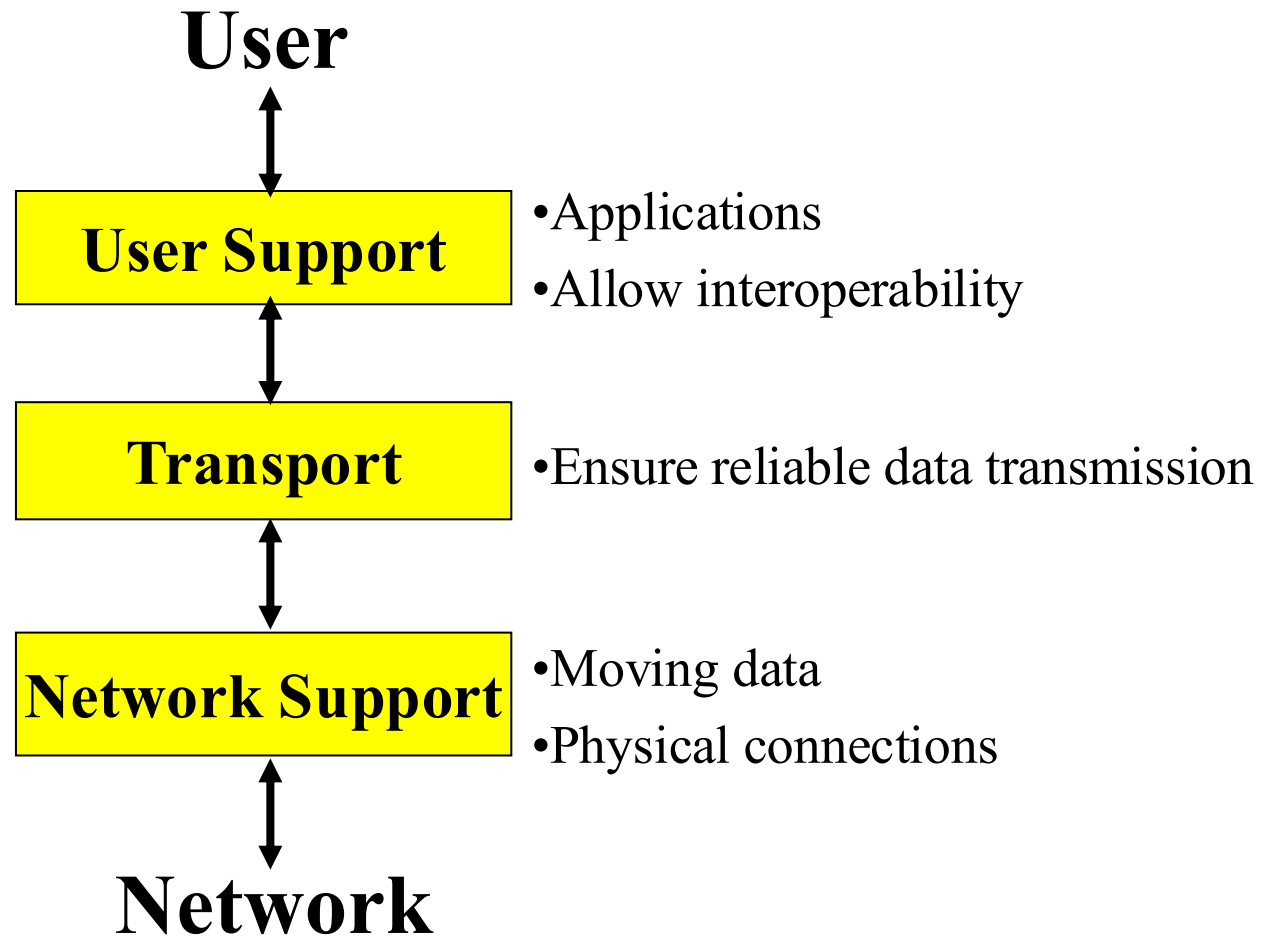
What Actually Happens

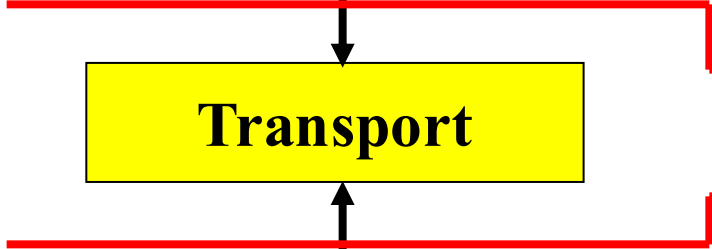
- Communication takes place thru many layers



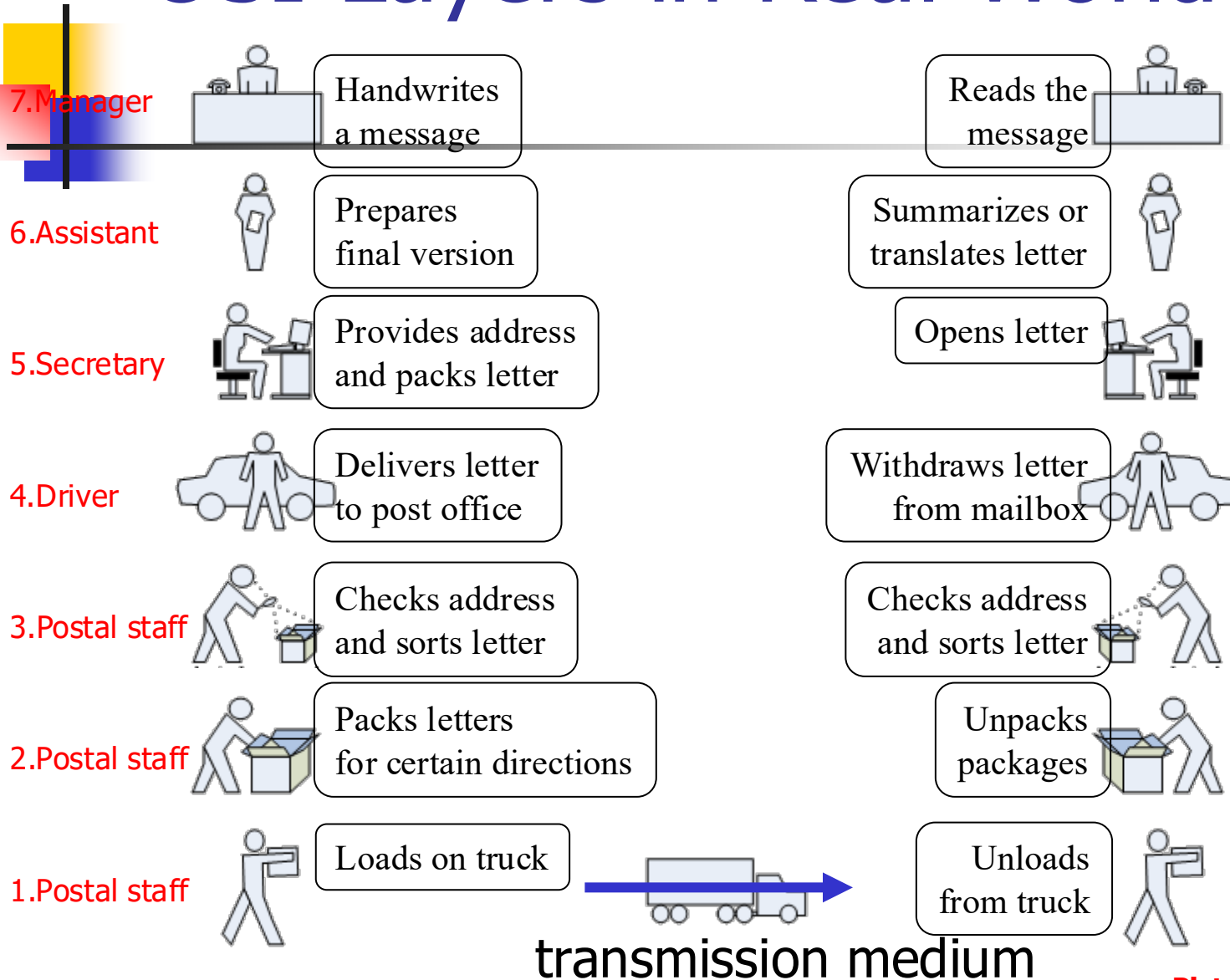


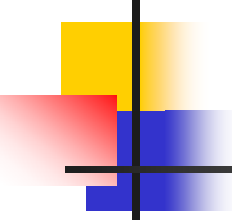
Layered (Software) Architecture





OSI Layers in Real World





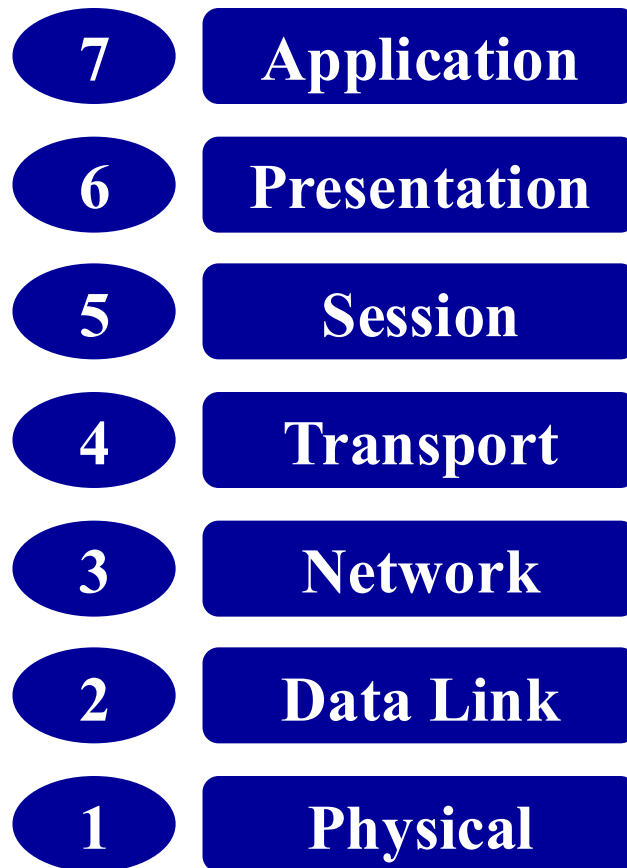
Old Protocol Stacks

Vendor	Stack
Novell Corporation	Netware
Banyan System Corporation	VINES
Apple Computer Corporation	AppleTalk
Digital Equipment Corporation	DECNET
IBM	SNA

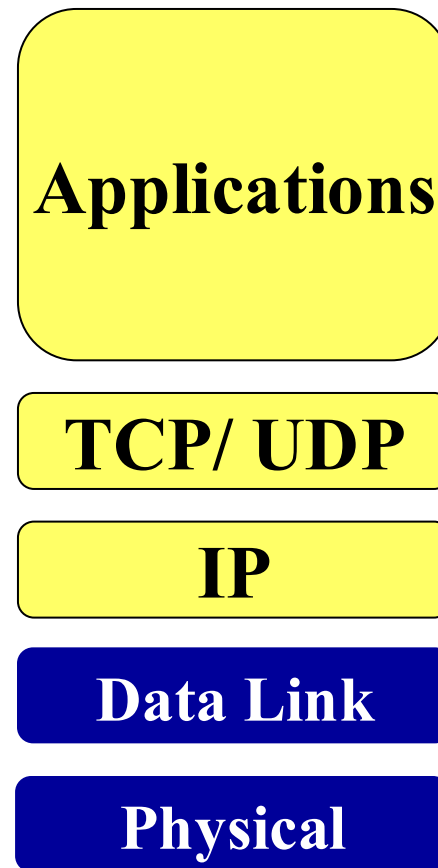
Now they are replaced by TCP/IP

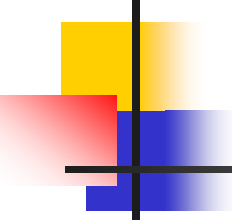
OSI model

“It’s just a model”



TCP/IP





TCP/IP Protocol Suite

(Internet Model)

5

Applications

User service and interface

4

Transport

Process delivery + Error (TCP/UDP)
Reliable end-to-end (whole message)

3

Network

Move packets from source to destination
Packet end-to-end (across network)

2

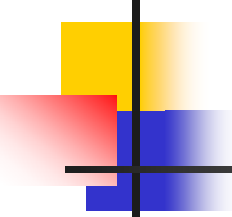
Data Link

Provide frames
Node-to-node (same network segment)

1

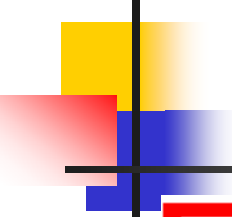
Physical

Transmission bit streams
(mechanical and electrical spec)



HW

1. Draw “The Lord of the Rings: Lighting the Beacons” in terms of TCP/IP protocol suite
 - Explain
 - Actors for each layer
 - The protocol used
 - Who is the sender and receiver?
 - How can the delay occur?
 - How about the acknowledgment from Rohan?

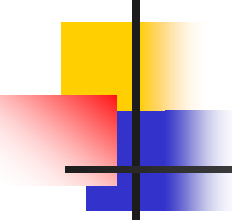


Outline



Network Model

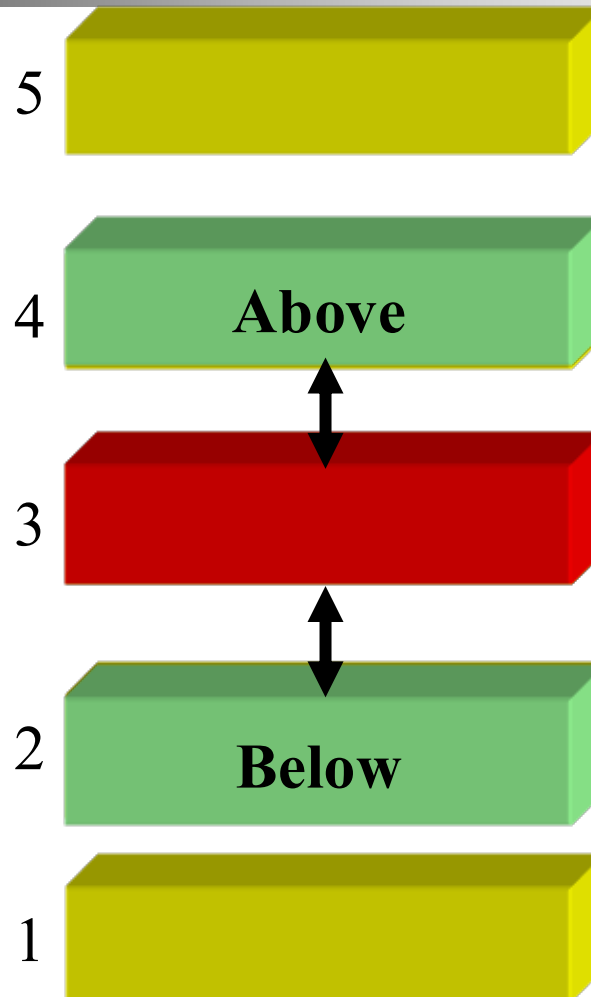
- **Layering Concept**
- Techniques protocols use



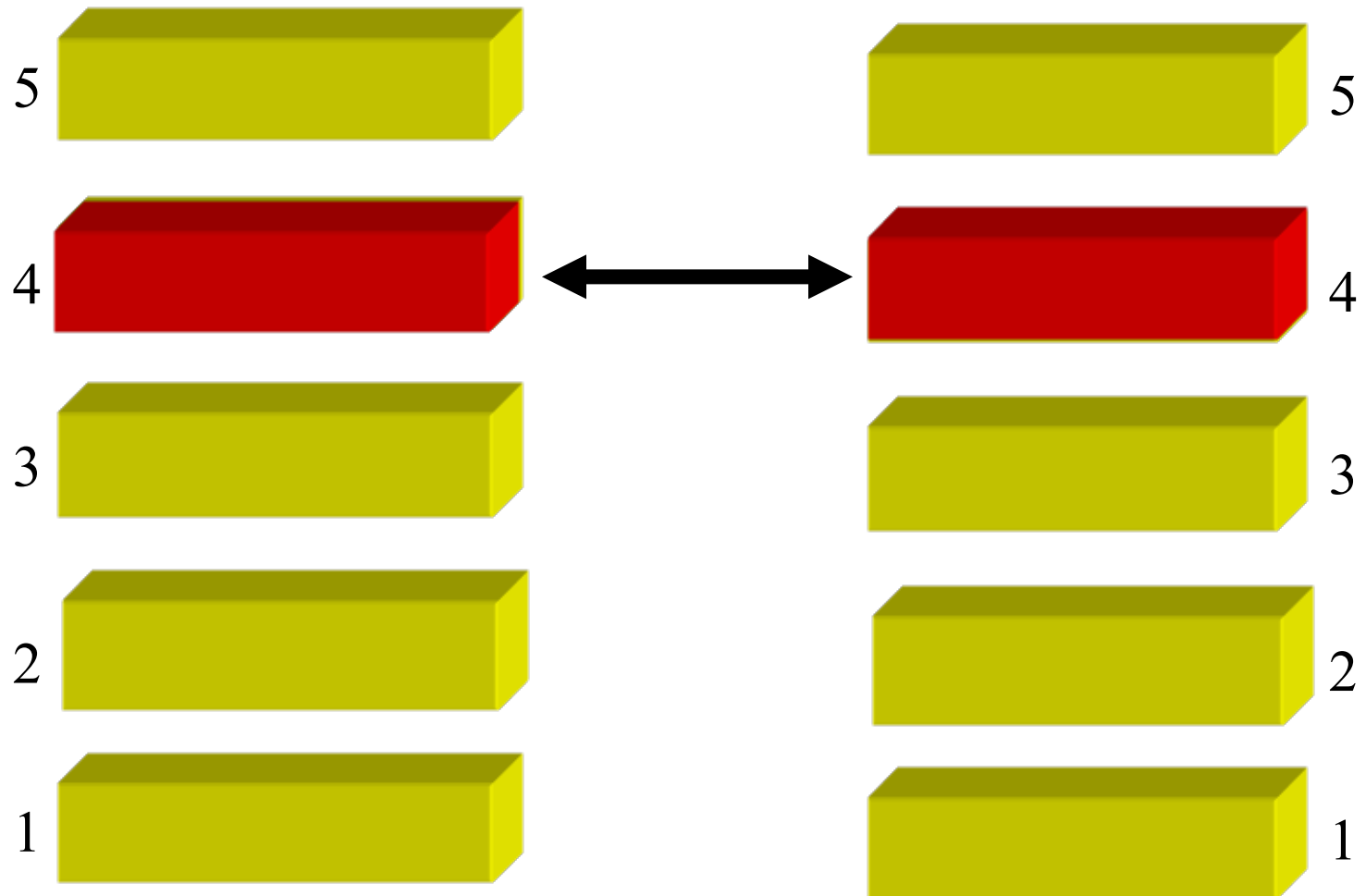
Layering Concept

1. Interfaces between layers (Physical)
2. Peer-to-Peer process (Logical)

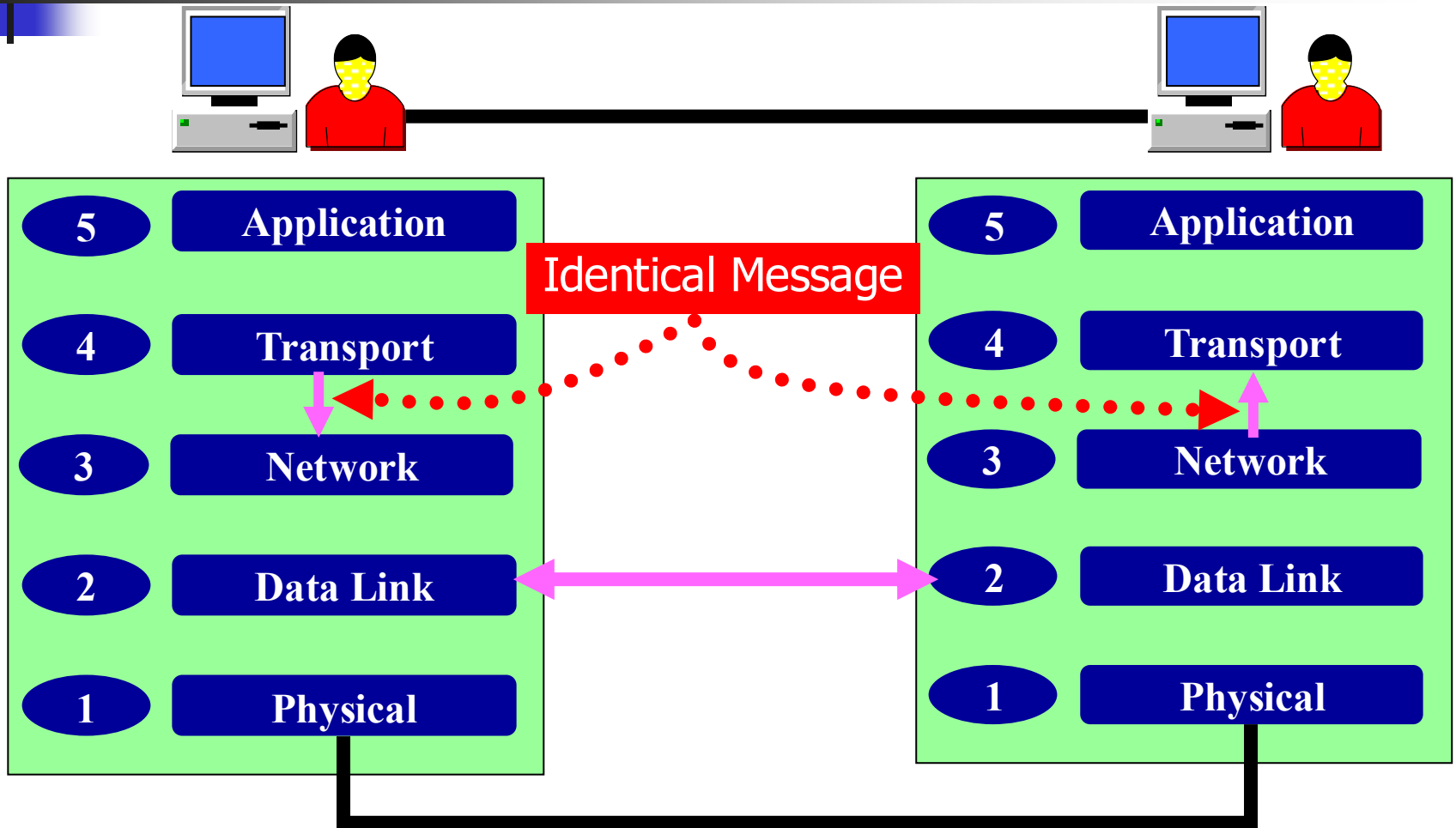
1. Interfaces between layers (Physical)



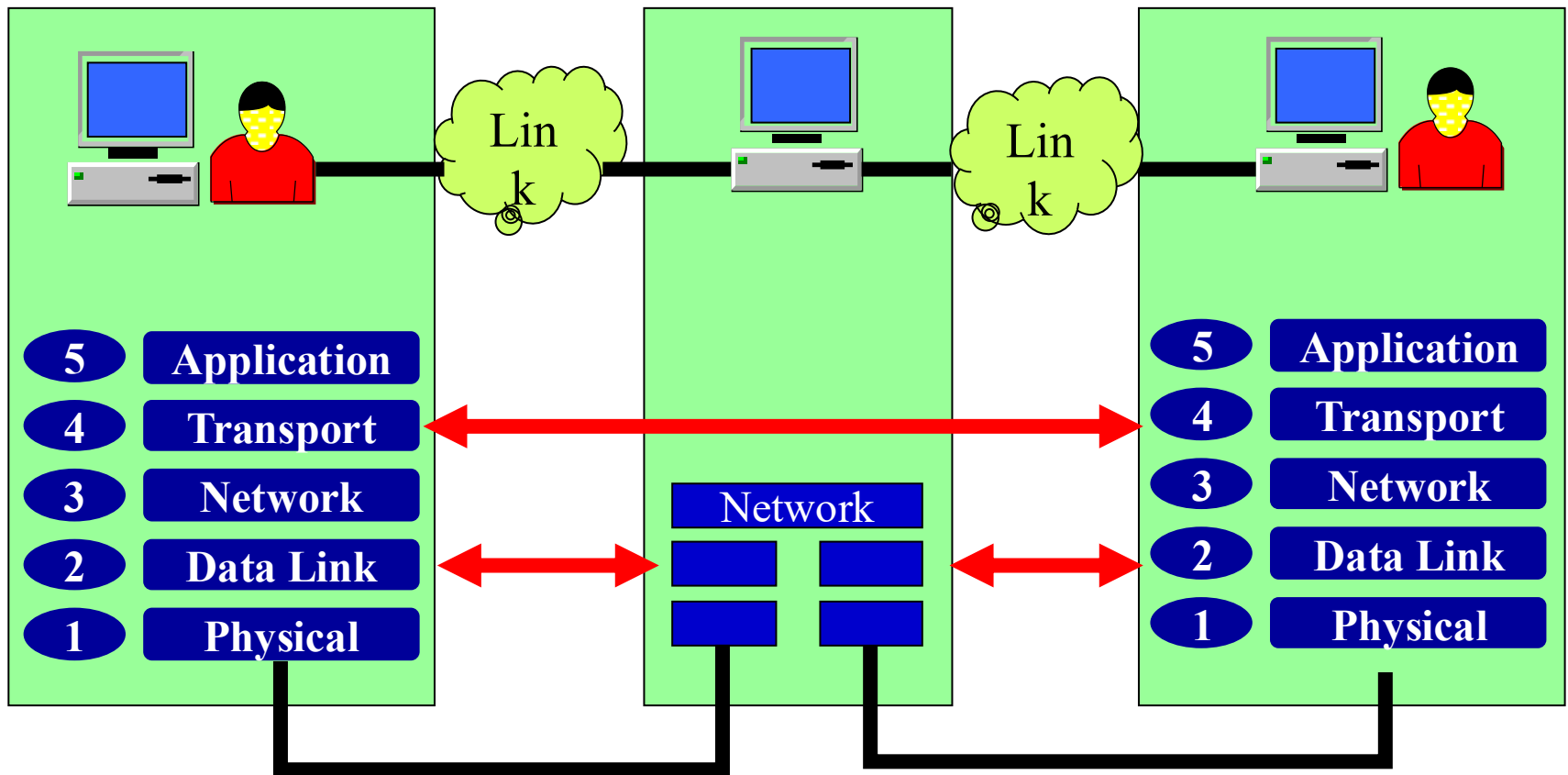
2. Peer-to-Peer process (Logical)



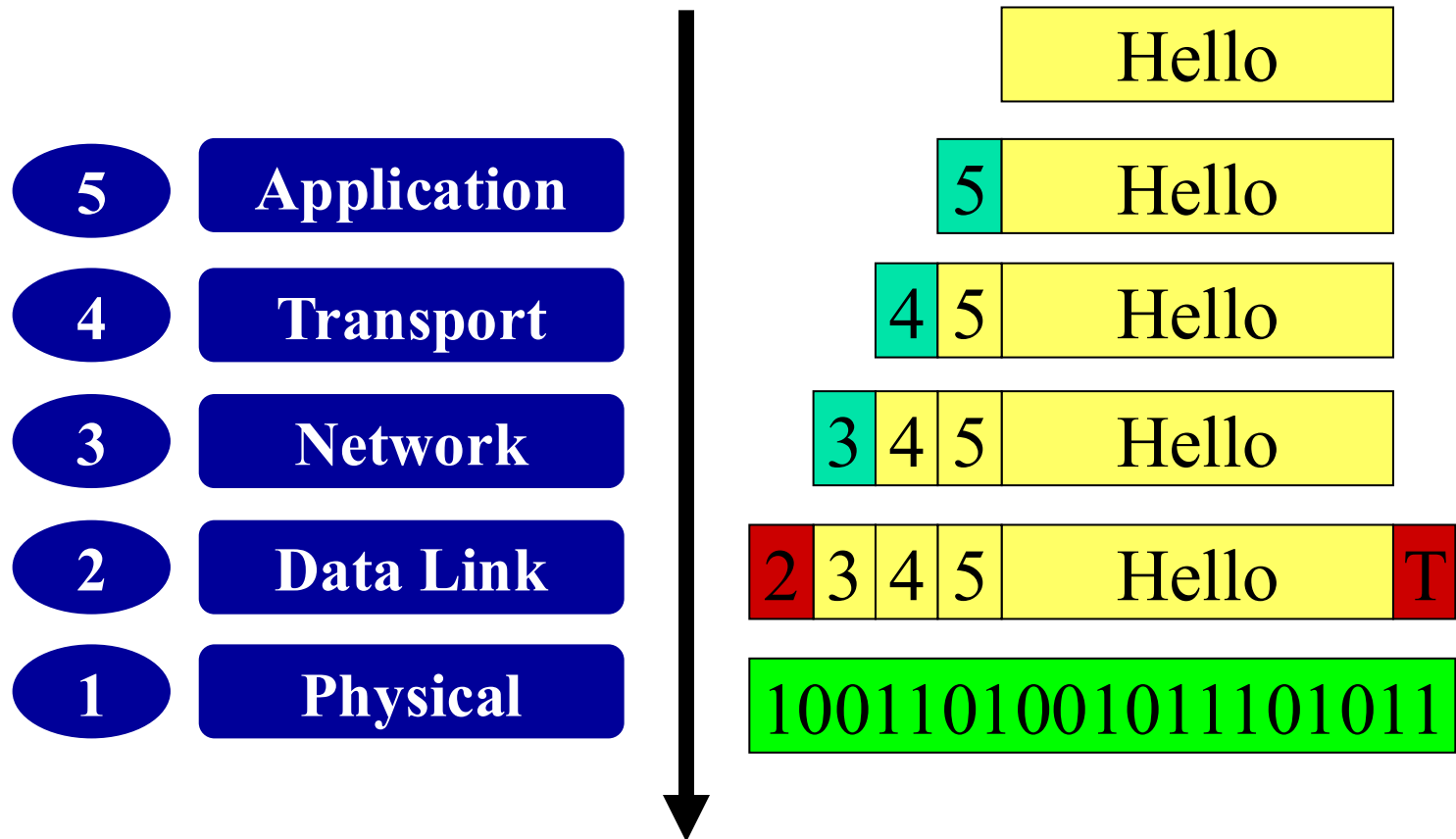
Direct connection



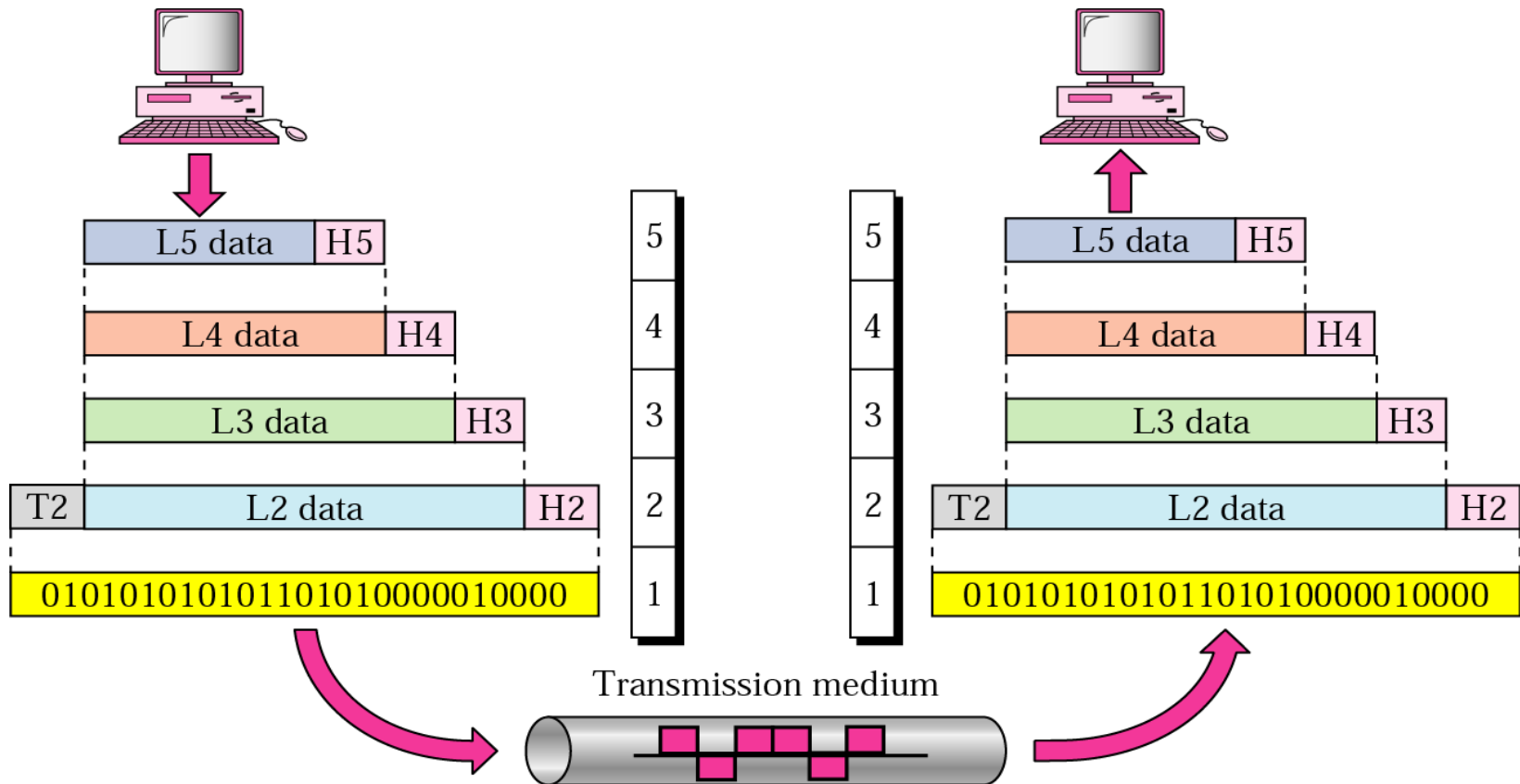
Connection via intermediate nodes

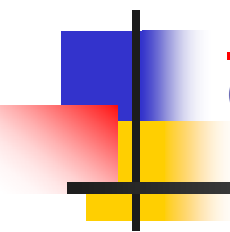


Data Flow in a station



Data Flow





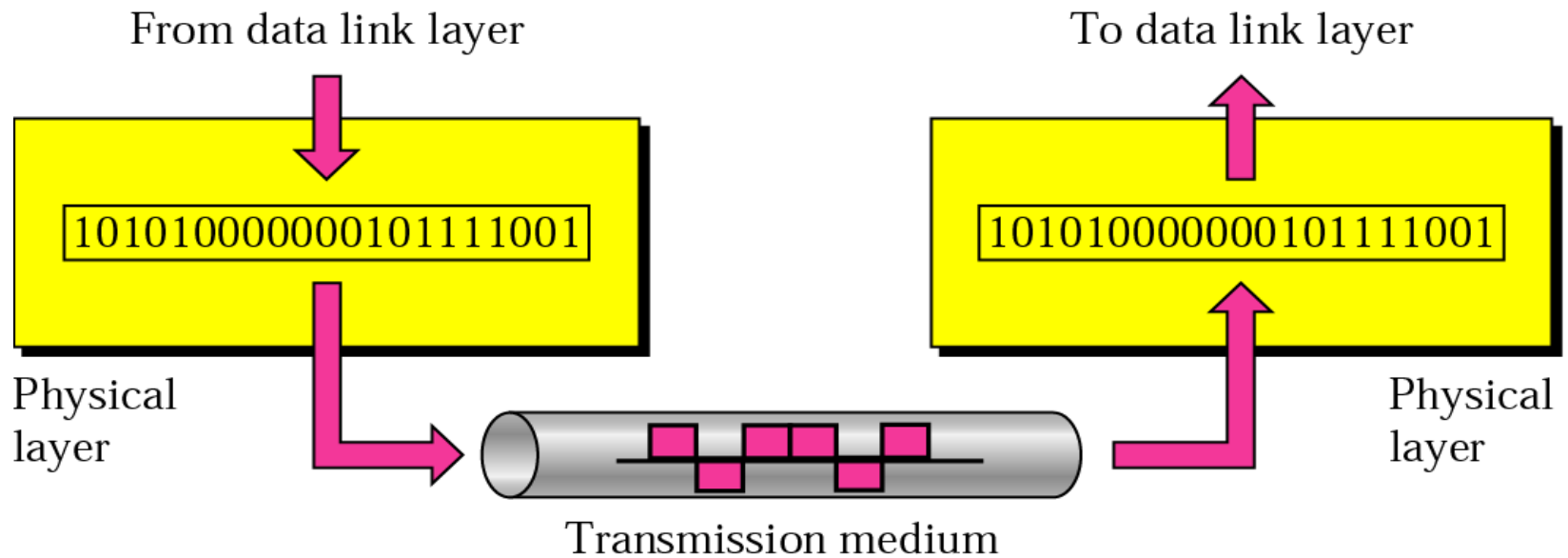
OSI Model and data flow

(03_03-DataFlow.swf)

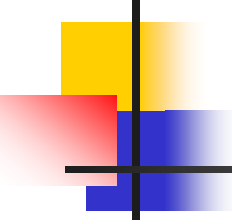
To run flash player

https://ruffle.rs/downloads?utm_source=chatgpt.com

1. Physical Layer



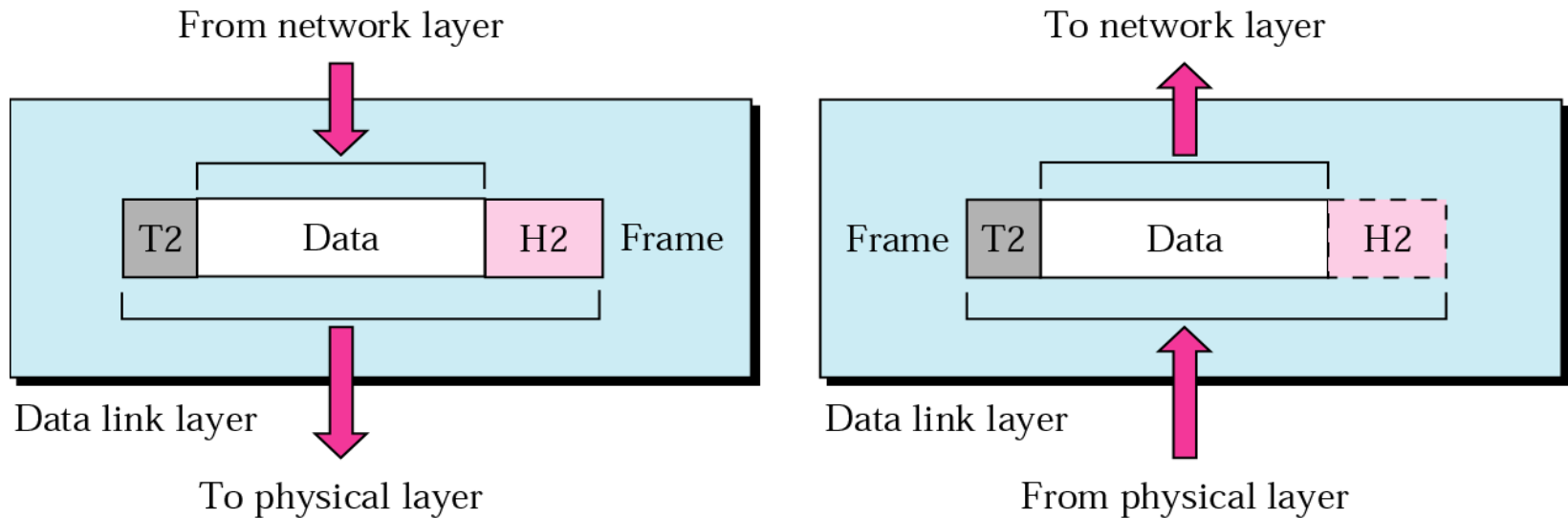
Transmitting individual bits from one node to the next.



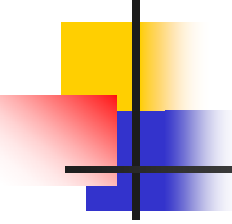
1. Physical Layer

- Physical characteristics of interface
- Stream of Bits (e.g., 001010100100)
- Line config. (e.g., point-to-point)
- Topology (e.g., bus, star)
- Transmission mode (e.g., half-duplex)

2. Data Link Layer



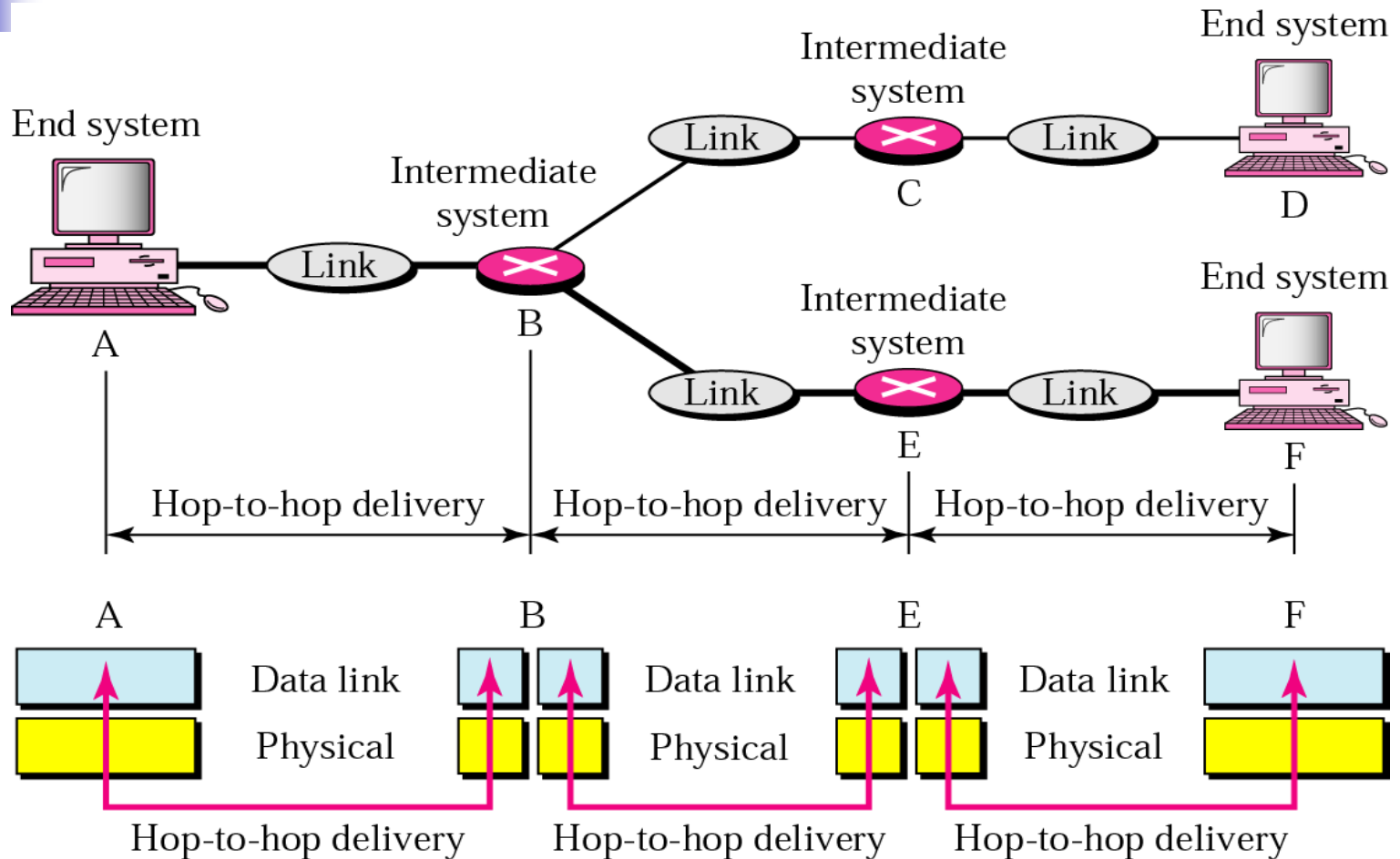
Transmitting frames from one node to the next.



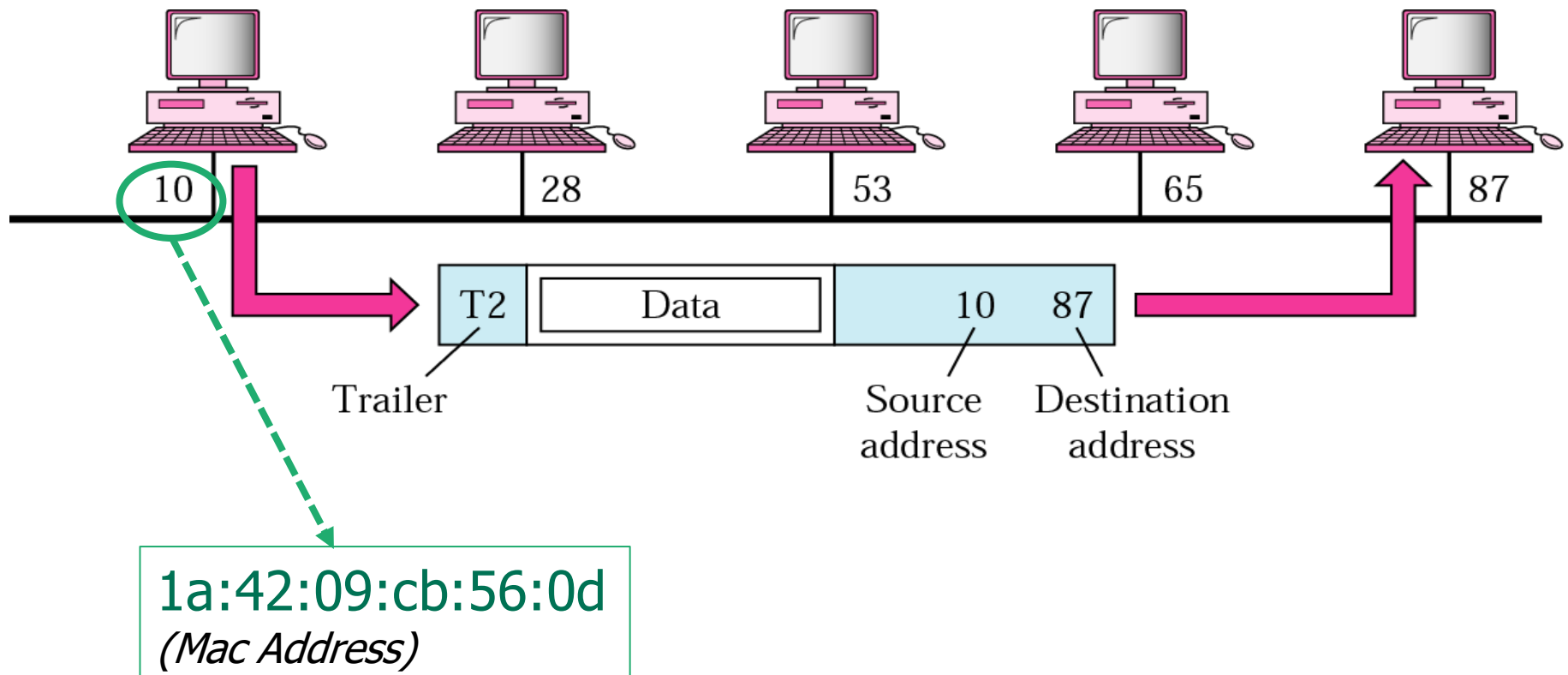
2. Data Link Layer

- Physical addressing
- Access control
- Error control
- **Node-to-Node delivery** (same network)

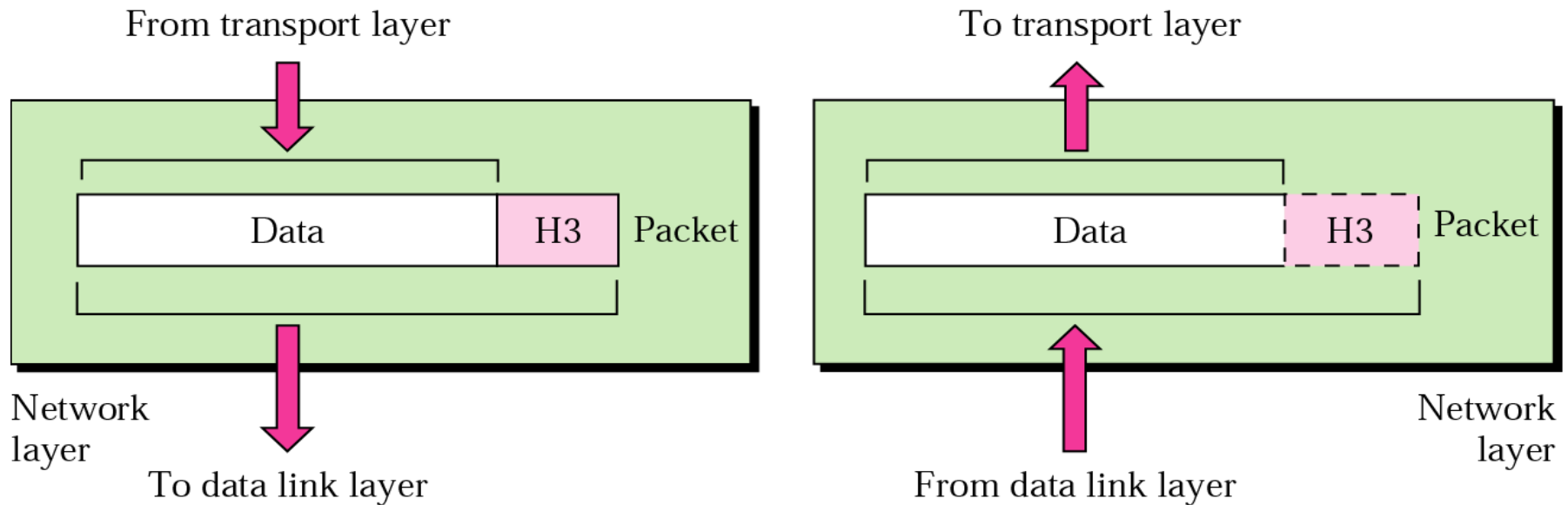
Node-to-node delivery



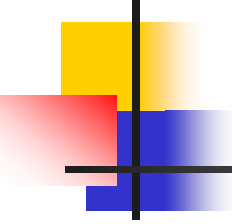
2. Data Link Layer Example



3. Network Layer



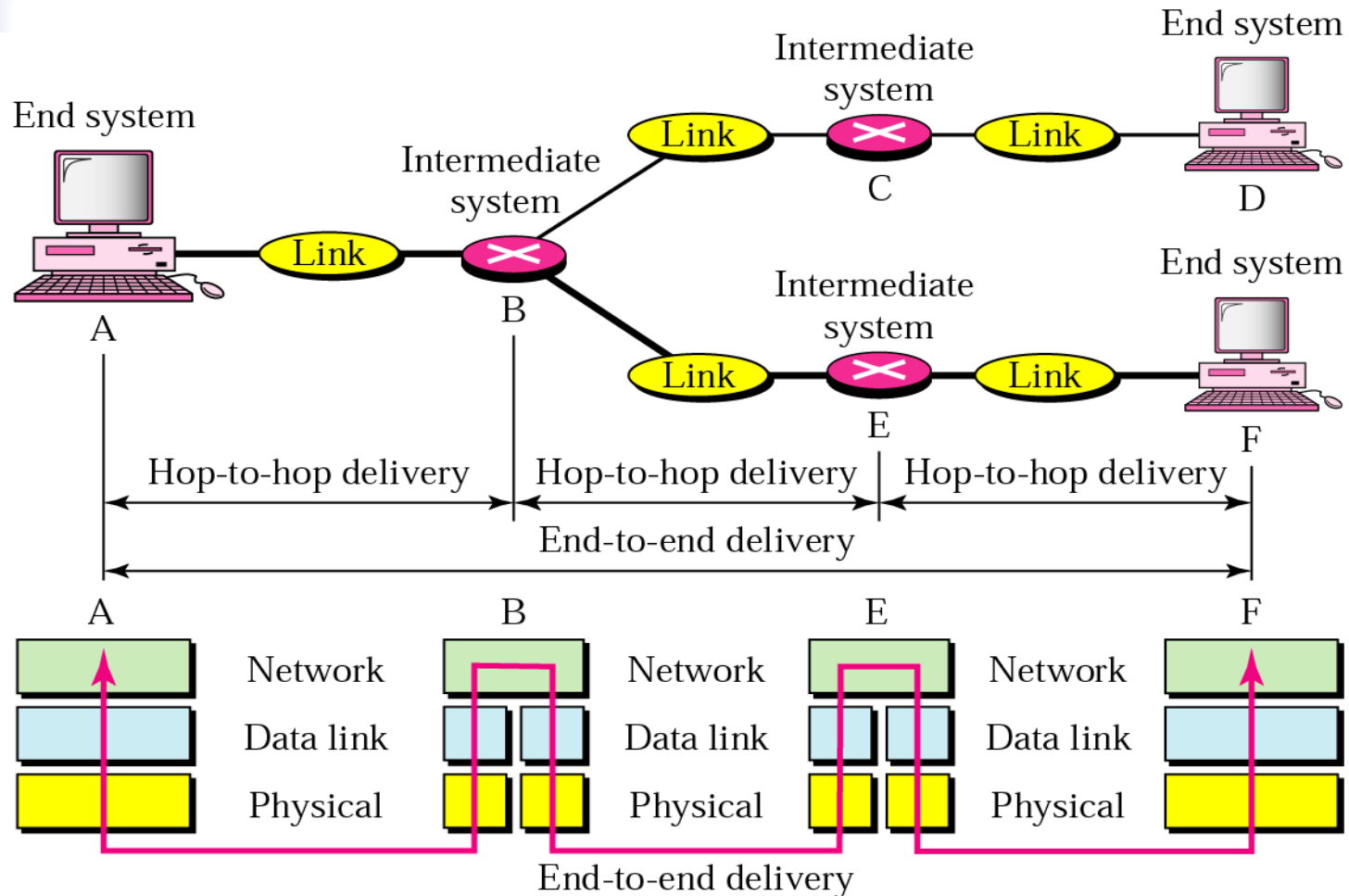
Delivery of packets from the source to the final destination

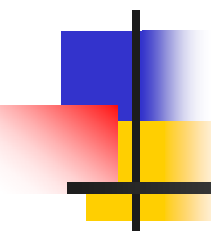


3. Network Layer

- Logical Address
(e.g., IP address:158.108.33.66)
- Routing (router, gateway)

Source-to-destination delivery

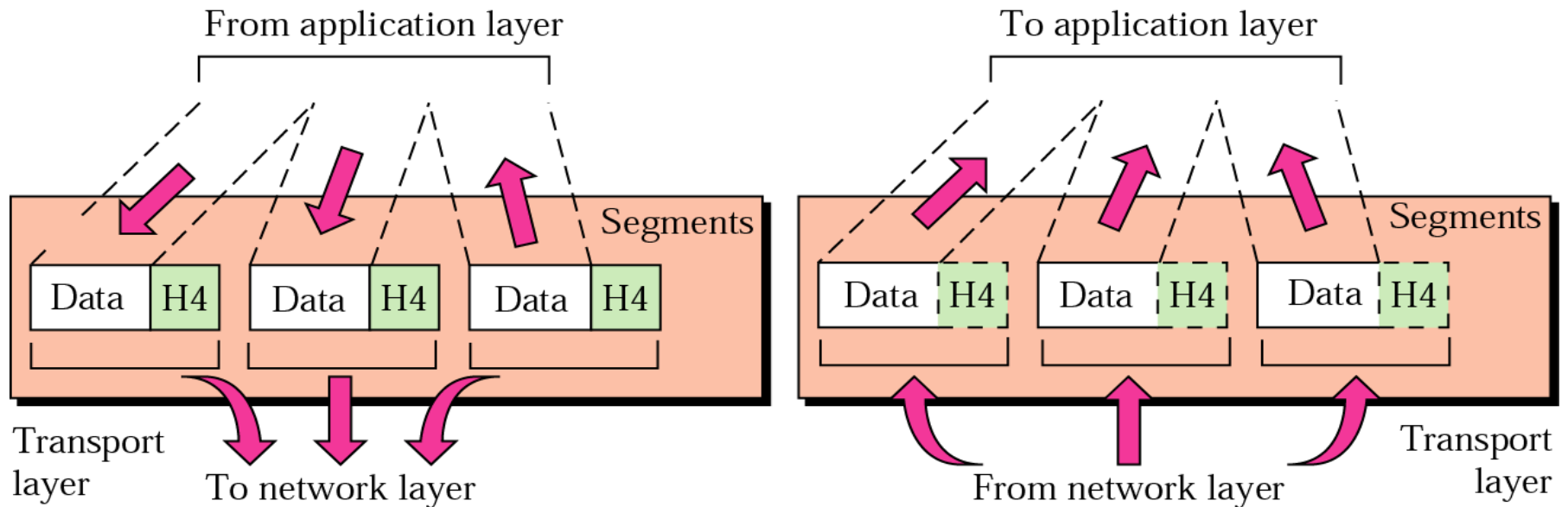




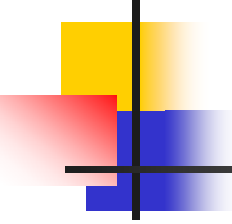
3. Network Layer Example

03_08-NetworkLayer.swf

4. Transport Layer



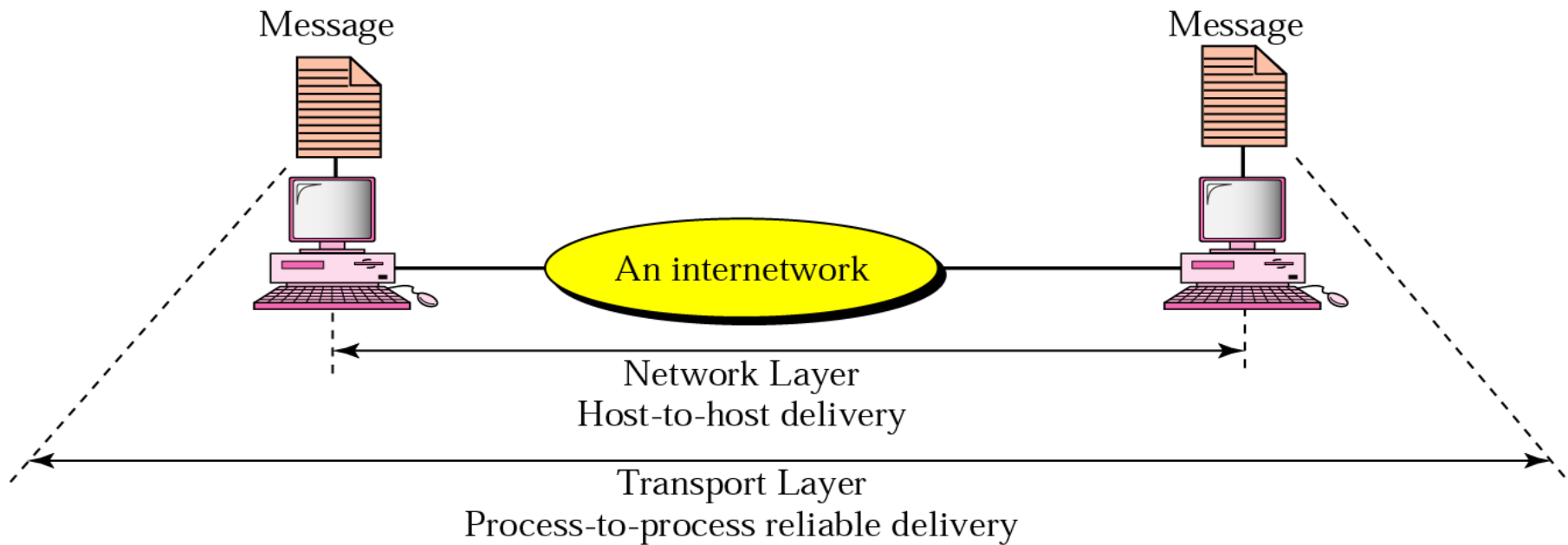
Delivery of a message from one process to another



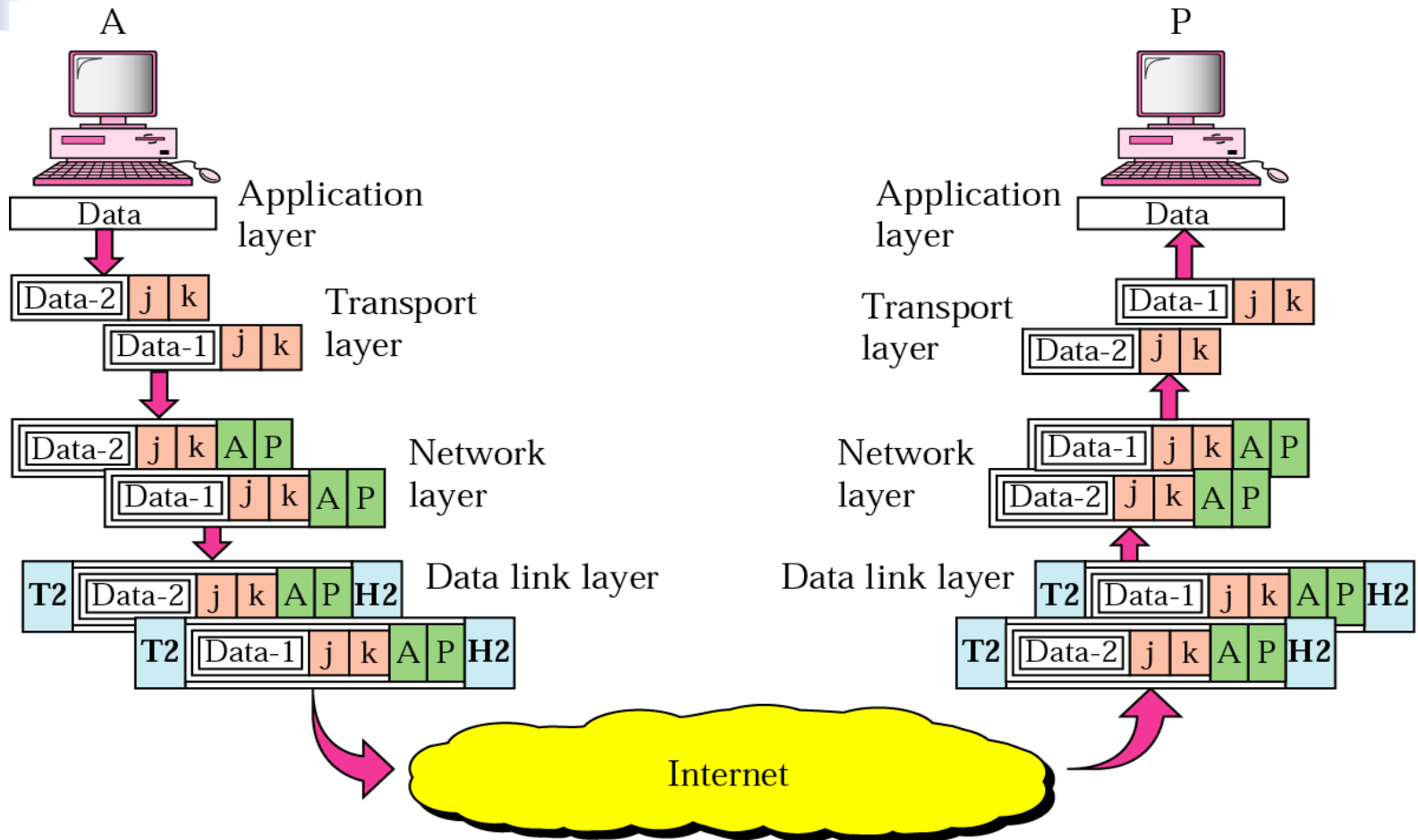
4. Transport Layer

- Service-point addressing (port number)
- Segmentation and assembly
- Flow and error control
- End-to-end delivery (across network)

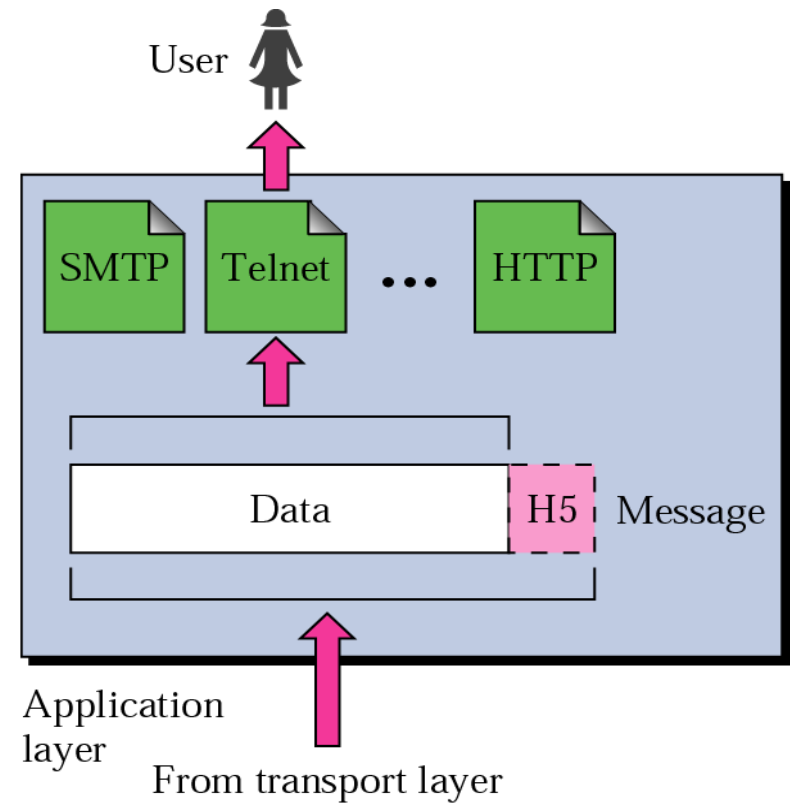
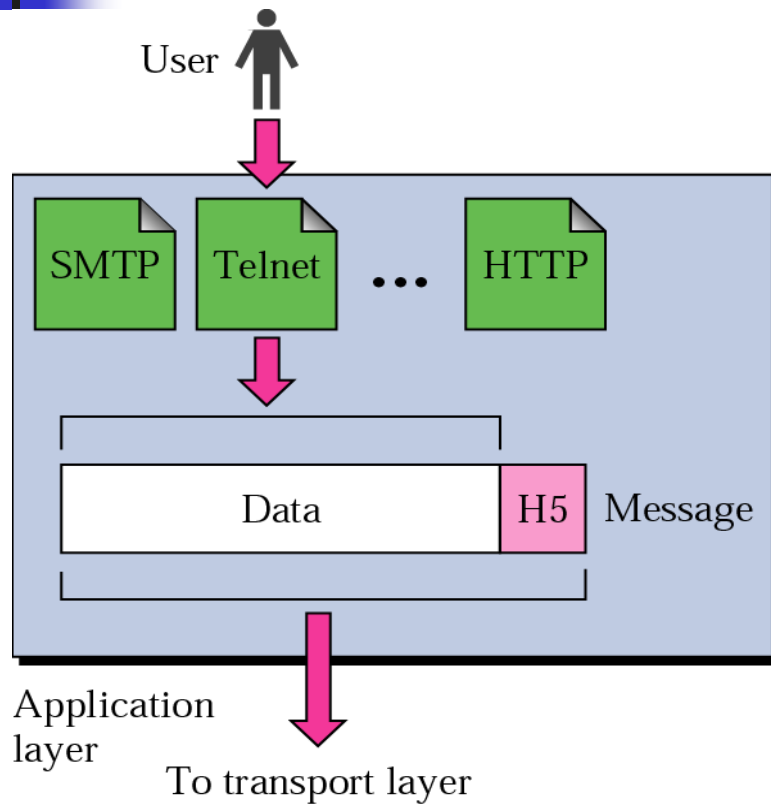
Process-to-process delivery



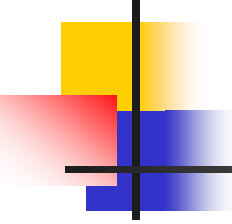
Process-to-process delivery



5. Application Layer

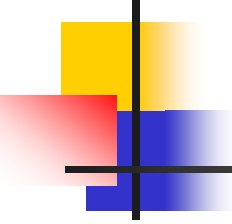


Providing services to the user



5. Application Layer

- User interfaces
- Service supports (e.g., mail, FTP, Game)



TCP/IP Protocol Suite

(Internet Model)

5

Applications

User service and interface

4

Transport

Process delivery + Error (TCP/UDP)
Reliable end-to-end (whole message)

3

Network

Move packets from source to destination
Packet end-to-end (across network)

2

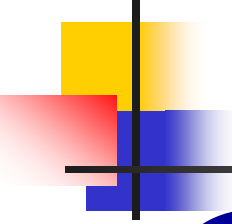
Data Link

Provide frames
Node-to-node (same network segment)

1

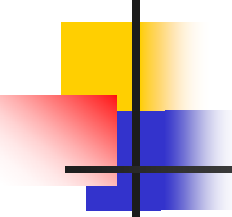
Physical

Transmission bit streams
(mechanical and electrical spec)



OSI Model

7	Application	User service
6	Presentation	Translate format, encrypt
5	Session	Session manage, checkpoints
4	Transport	Reliable end-to-end (whole message)
3	Network	Packet end-to-end (across network)
2	Data Link	Node-to-node (same network segment)
1	Physical	Physical



Outline

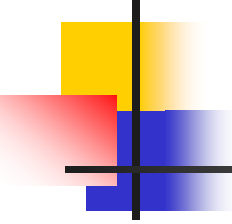


Network Model



Layering Concept

- **Techniques protocols use**



Techniques Protocols Use

1. Sequencing Number

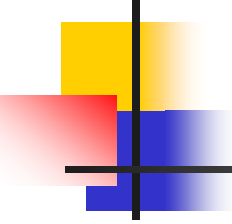
- for out-of-order delivery
- to eliminate duplicate packets

2. Retransmission lost packets

3. Avoiding replay caused by excessive delay

4. Flow control to prevent data overrun

5. Mechanisms to avoid network congestion

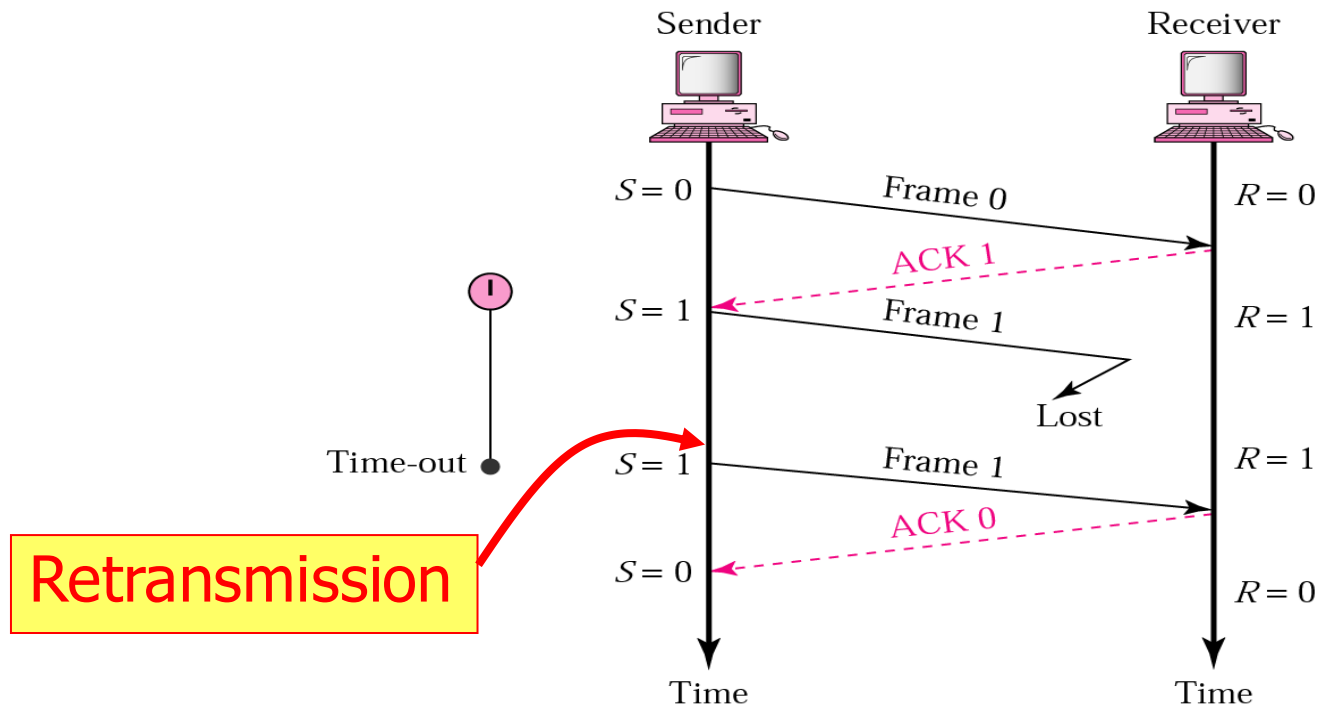


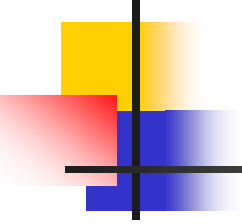
1. Sequencing Number

- Sender adds sequence no. in the packet
- For out-of-order delivery
 - sorting in the list
- To eliminate duplicate packets
 - discard the new one

2. Retransmission lost packets

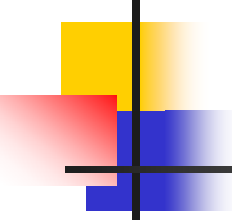
- To guarantee reliable transfer
 - Positive **acknowledgement (ACK)**
 - Retransmission





3. Avoiding replay caused by excessive delay

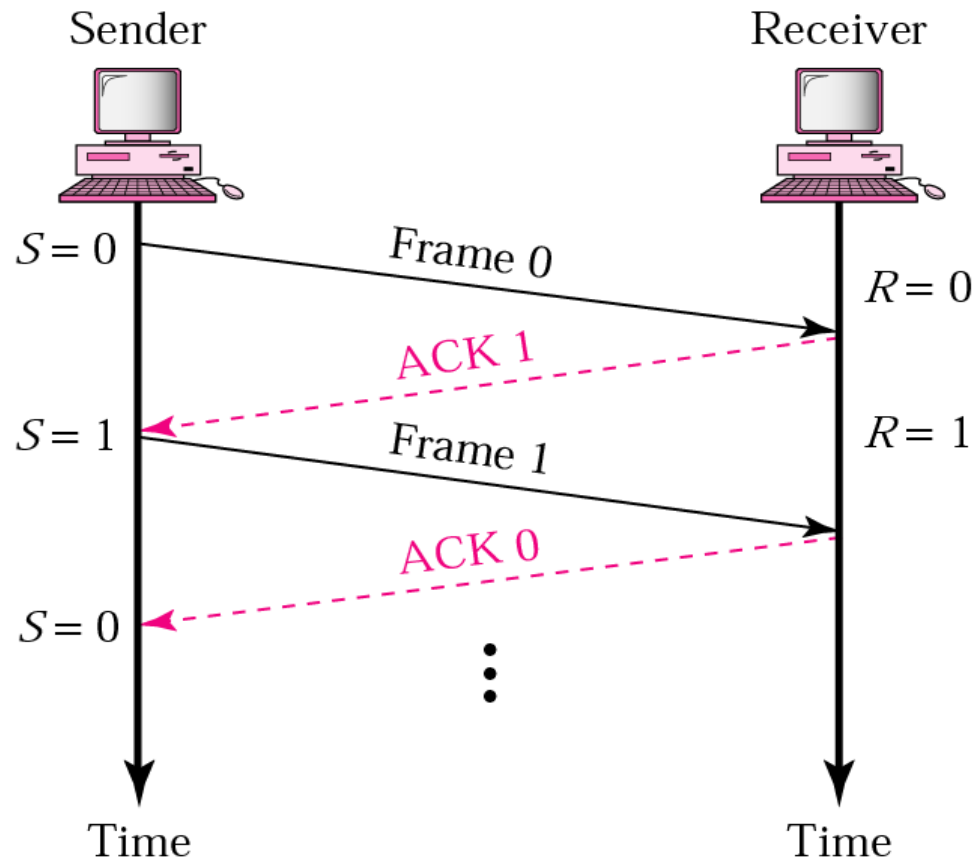
- A packet from previous conversation comes late
- Just discard the old packet
- Detected by a unique ID (e.g. time)



4. Flow control to prevent data overrun

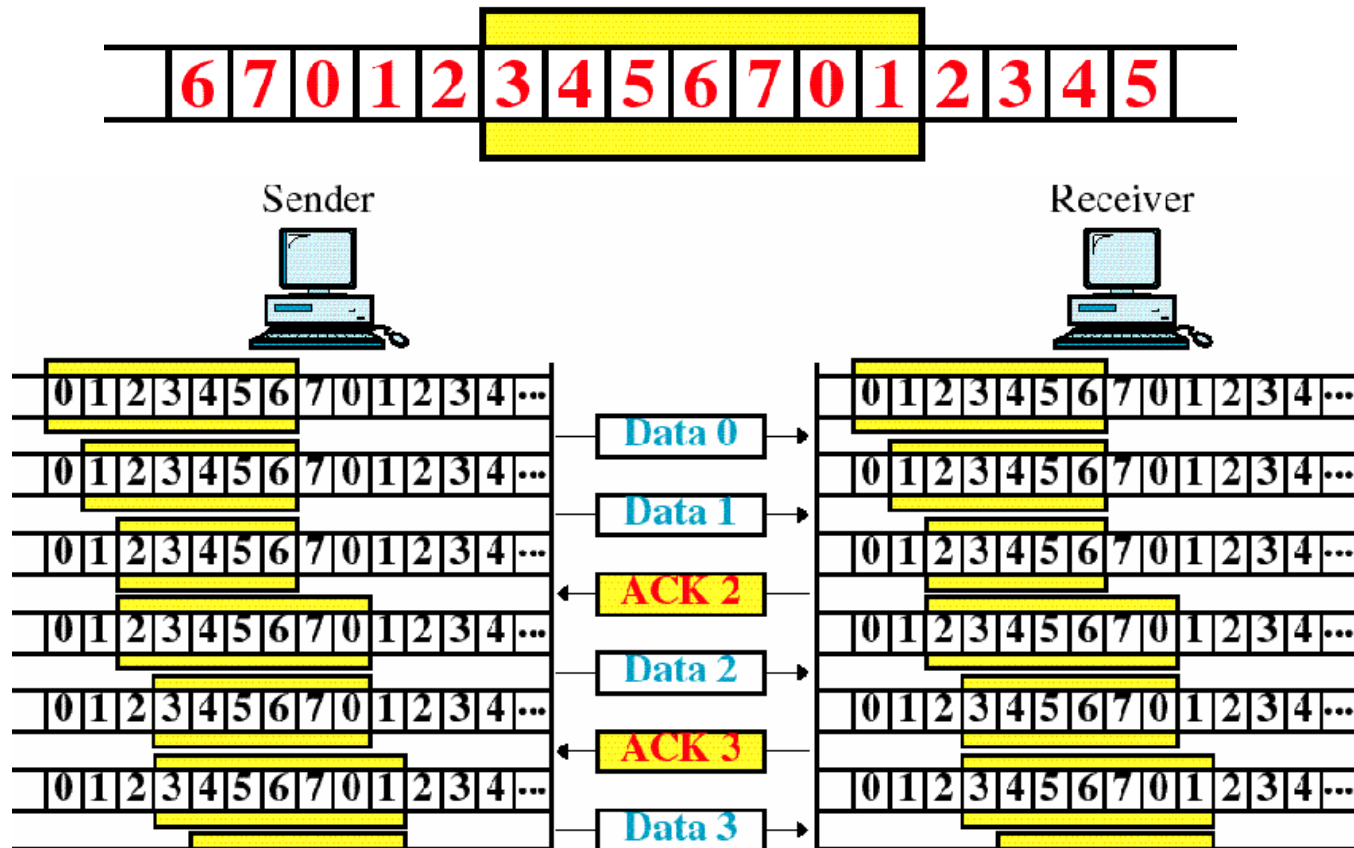
- Sender / receiver may operate at different speed
→ Data is lost
- Flow control mechanism
 - Stop-and-go (Stop-and-wait)
 - Sliding window

Stop-and-go (Stop-and-wait)

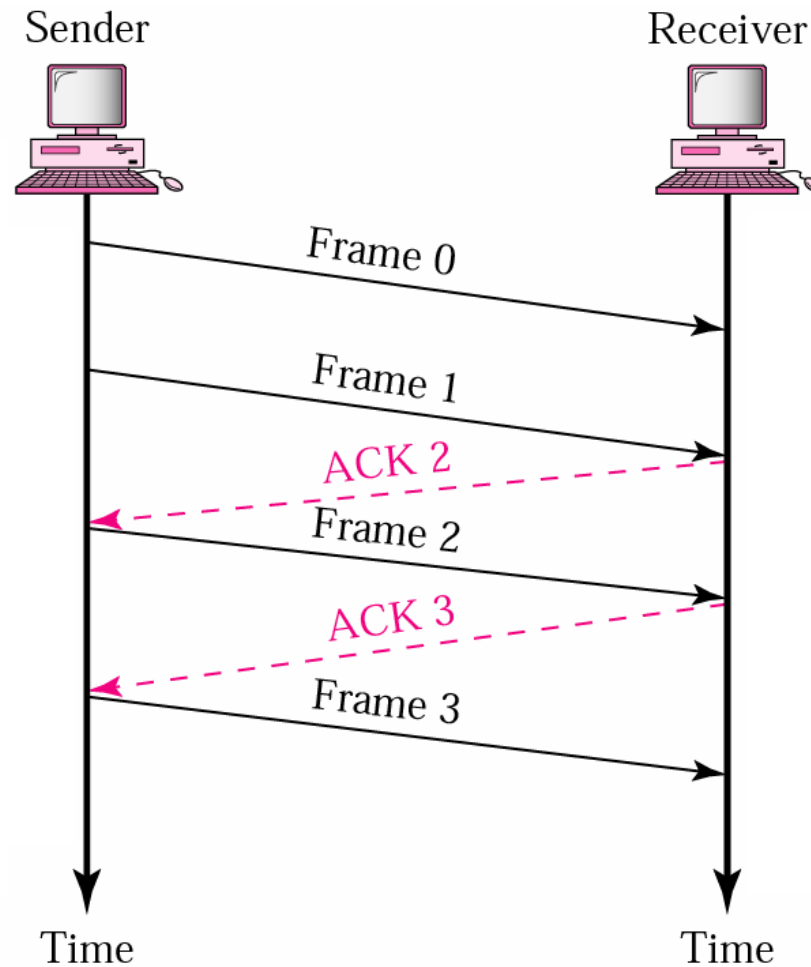


Sliding window

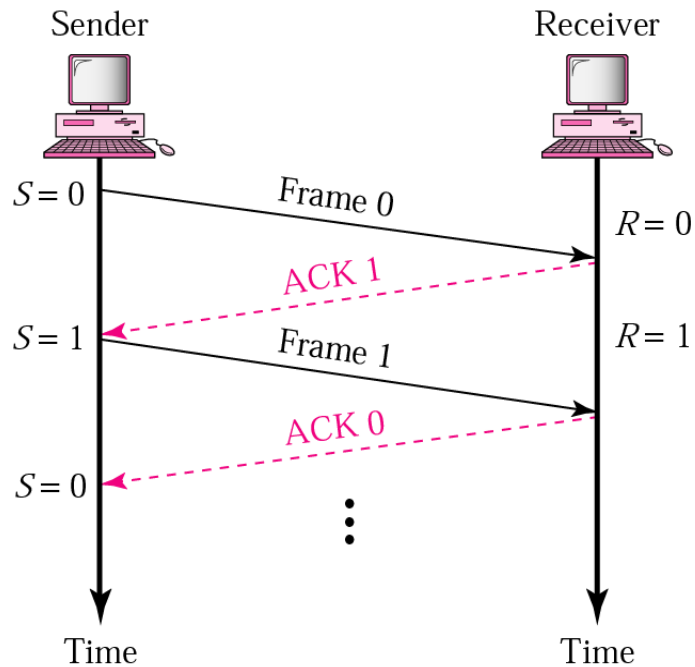
Window



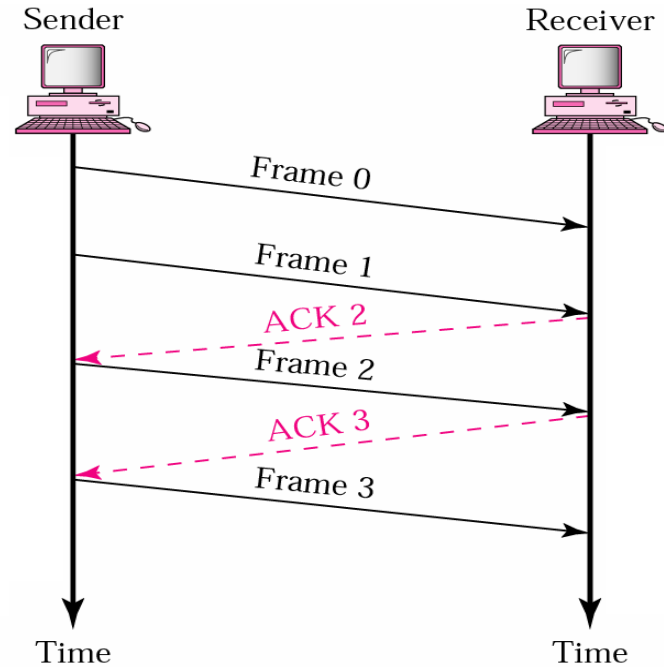
Sliding window



Comparison



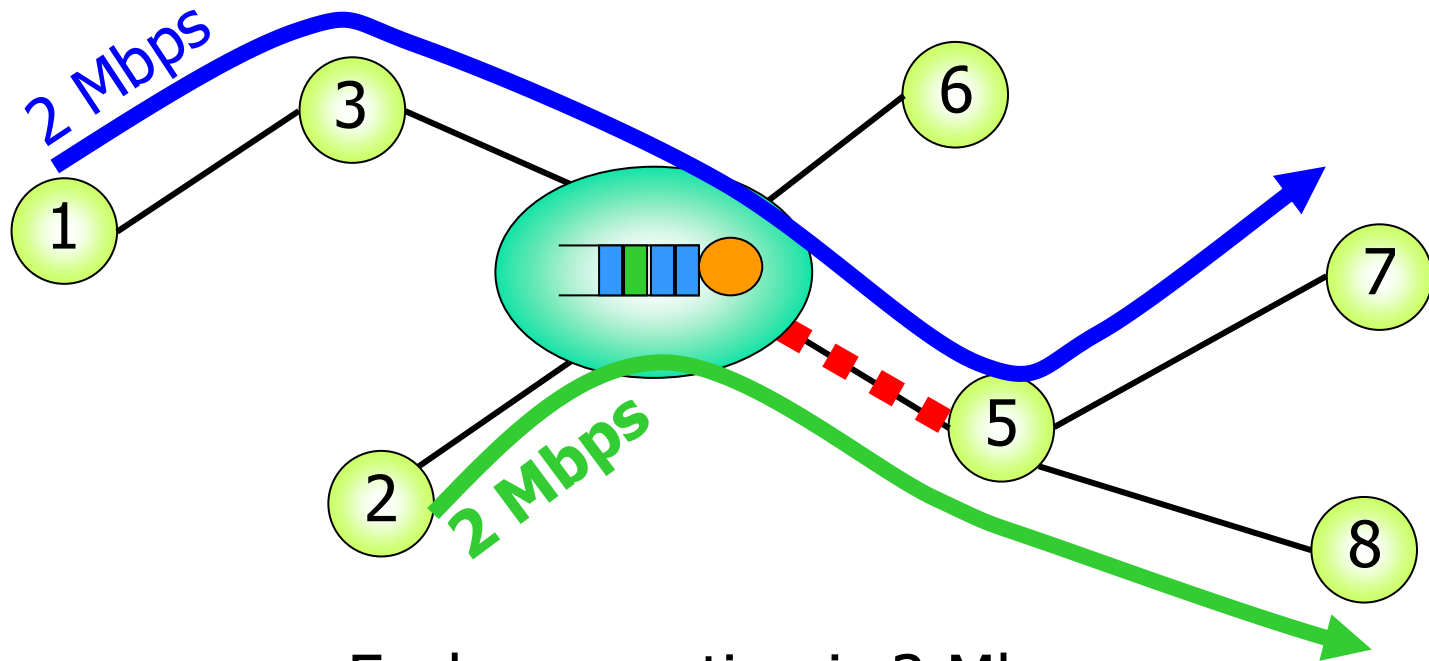
Stop-and-go



Sliding window

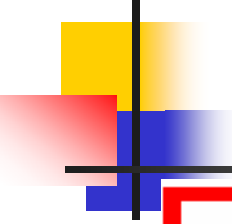
Can you compare throughput ?

5. Network congestion



Each connection is 3 Mbps

To avoid congestion → control the incoming rate



Summary



Network Model

- OSI Model
- TCP/IP protocol suite



Layering Concept

- Data flow
- functions of each layer



Techniques protocols use

- Seq.no. / Retransmission / Congestion